Solar Panel Cooling + Cleaning System: Project Record

Compiled from our conversation, 10 October 2026. This is an organised record of everything decided so far: the component review, the design, wiring, lab parts, and the 3D printed parts.

1. Project in one paragraph

An automated system that cools a solar panel with a water spray when it gets hot, and cleans it with a wiper carriage when dust lowers its output. An ESP32 reads sensors and controls a pump and a motor. A second, identical panel is left untouched as a control, so the benefit can be measured and written up. Team context: first-year multidisciplinary project (3 CSE, 1 Mechatronics, 1 EEE) aiming for a working application, a paper and possibly a patent.

2. How the chat went (timeline)

1. Reviewed the teammate's component list and found gaps and mistakes.
2. Produced a corrected total parts list.
3. Produced the full design: mindmap, physical layout, power circuit and ESP32 wiring.
4. Turned the parts list into an Excel sheet.
5. Decided a Raspberry Pi is not needed.
6. Checked every part against the college lab lists (MDP/IDP labs and IoT lab) and found replacements.
7. Made a lab pick-up list.
8. Wrote detailed connections and the mechanical design of the cleaning carriage.
9. Generated 3D printable parts (STL files) with a fit-test coupon.

3. Review of the teammate's list

Verdict: a decent start for electronics, but not enough to build the system. It had no solar panel, pump, nozzles, tank or cleaning mechanism, and several items were overkill or wrong.

Main problems found:

- **No solar panels, pump, nozzles, tank, wiper, rails or belt.**
- **ACS712 30A:** poor resolution for a panel producing about 1 A.
- **Voltage sensor module:** divides by 5, so a 21 V panel gives 4.2 V on a 3.3 V ESP32 pin.
- **DHT11:** inaccurate; and no sensor measured panel surface temperature (use DS18B20 probes on the panel back).
- **Jetson Nano / Raspberry Pi 5:** overkill and power hungry (5-12 W against a 20 W panel).
- **ESP8266:** only one analog pin.
- **Single 18650 cell:** cannot run a 12 V pump and motor; a 3S pack with BMS is needed.
- **No charge controller, fuses, diodes, capacitors, level or rain sensing, or limit switches.**
- **Redundant choices** (both DC motor and stepper drivers, three cameras): pick one of each.
- **Breadboard:** fine for testing only; final wiring should be on perfboard with screw terminals.

4. System design

What it does

1. **Sense:** panel temperature, voltage, current, sunlight level, rain, tank level.
2. **Cool:** above 45 C panel temperature, run the pump until it falls to 38 C (maximum 60 seconds, skipped in rain or with an empty tank).
3. **Clean:** when the soiling index stays below about 0.9 for 10 minutes, or on schedule, spray and run the carriage down and back.
4. **Log:** timestamped data to a dashboard.

Soiling index

(Panel A current divided by reference-cell current) now, relative to the same value right after the last clean. Only calculated in good light so clouds do not trigger it.

Architecture

- **Power:** two 12 V panels, charge controller, 3S 18650 pack with BMS, 12 V rail, LM2596 buck to 5 V.

- **Brain:** ESP32 with Wi-Fi.
- **Cooling:** tank, 12 V pump through a relay, spray bar with nozzles at the top of the panel.
- **Cleaning:** two 8 mm rods along the panel sides, a carriage on LM8UU bearings, a cross bar, pivoting wiper arms with a silicone squeegee, GT2 belt driven by a 12 V gear motor.
- **Control panel:** Panel B, untouched, into a dummy load.

5. Components, as finally decided

Power

- 2 x 12 V 10-20 W solar panels
- Charge controller with a Li-ion (3S, 12.6 V) setting, or a CN3791 module set to 3S
- 3 x 18650 cells, 3S holder, 3S BMS
- 2 x LM2596 buck converters
- Schottky diode, fuses (3 A and 5 A), master switch, two dummy loads (10 W resistor or 12 V bulb)

Controller and logging

- ESP32-S2-DevKitM-1 from the lab (ESP32-S3 also possible, but the pin map would need remapping)
- ESP32-CAM (optional)
- Time from Wi-Fi (NTP) instead of a DS3231 module; cloud logging instead of a microSD module

Sensors

- 2 x ACS712 30 A (current), with resistor dividers
- Voltage: own resistor dividers (the EPRO module is not used)
- 3 x DS18B20 (panel temperature), DHT22 (ambient)
- Reference mini solar panel (0.45 W) across a 10 ohm resistor as the sunlight reference
- Soil moisture module as a rain detector
- Optical liquid level sensor (or ultrasonic) for the tank
- IR sensor modules or push buttons for end stops
- Optional thermal sensor (AMG8833 or MLX90614)

Actuators

- 12 V mini pump, relay module (5 V, 2-channel), flyback diode
- 25GA-370 12 V gear motor (60 RPM version), L298N driver
- Optional MG995 servo and MOSFET module

Water and cleaning hardware

- Spray nozzles, 12 mm spray bar tube, tubing, check valve, inline filter, water tank
- 2 x 8 mm smooth rods, 4 x LM8UU bearings, 8 mm aluminium cross bar
- GT2 belt, 20-tooth pulleys, silicone squeegee strip, 15 x 15 aluminium angle, small springs or rubber bands

Electrical supplies, frame and tools

- Capacitors (220 uF or 1000 uF, and 100 nF), resistors (4.7k, 10k, 100k), 1N4007 diodes
- Power wire, hookup wire, terminals, XT60 connectors, perfboard, IP65 or steel enclosure, heat shrink
- Tilt stand, fasteners
- Multimeter, soldering kit, wire stripper, pliers, jumper wires, breadboard, screwdriver set, hot glue gun

6. Corrections made during the chat

- **INA226 changed to INA219** early on, because the common INA226 module saturates at about 0.8 A. Later, with the lab parts, current sensing moved to the **ACS712 plus dividers** (buy the INA219 later if you need paper-quality accuracy).
- **V-slot wheels replaced** by 8 mm rods with LM8UU bearings.
- **ESP32 WROOM replaced** by the lab's ESP32-S2 (or S3).
- **LCD/OLED:** the LCD backpack pulls I2C up to 5 V, which can stress ESP32 pins; the display is optional because the dashboard shows the data.
- **EPRO voltage modules** are not needed, because their divide-by-5 saturates the ESP32; use the 100k / 10k divider.
- **Printed carriage blocks:** two are needed, not one.

7. Raspberry Pi: not needed

The ESP32 does everything the design requires. A Pi only makes sense for on-device image-based dust detection. Alternatives: the soiling index (no camera), or an ESP32-CAM sending photos to a laptop or the cloud. A Pi's power draw would also have to be subtracted from the net-gain numbers.

8. Wiring

All grounds are joined into one common GND.

Power chain

- Panel A (+) to 5 A fuse, then ACS712-A IP+.
- ACS712-A IP- to the Schottky diode (anode in), diode cathode to charge controller PV+.
- Panel A (-) to charge controller PV- and the GND rail.
- Charge controller battery terminals to BMS P+ / P-.
- BMS B-, B1, B2, B+ to the three cells in series.
- Charge controller load output (or BMS P+) to the master switch, then a 5 A fuse, then the 12 V rail.
- 12 V rail feeds the LM2596 input, the L298N 12 V terminal, and the relay COM.
- LM2596 output becomes the 5 V rail. **Set it to exactly 5.0 V with a multimeter before connecting anything.**
- 5 V rail feeds the ESP32 5V pin, relay VCC, both ACS712 VCC pins, and the L298N +5V pin (remove the L298N 5 V jumper).
- Panel B (+) to a 3 A fuse, ACS712-B IP+, IP- to the dummy load, back to Panel B (-).

Charger caution: a lead-acid PWM controller charges to about 14.4 V, too high for 3 Li-ion cells (12.6 V maximum). Use a controller with a Li-ion setting, or a CN3791 set to 3S. Do not rely on the BMS alone.

ESP32-S2-DevKitM-1 pin map

- **GPIO 1:** Panel A voltage. Panel (+) to 100k, node to GPIO 1, node to 10k to GND, 100 nF across the 10k. Scale factor 11.
- **GPIO 2:** Panel A current. ACS712-A OUT to 4.7k, node to GPIO 2, node to 10k to GND, 100 nF across the 10k.
- **GPIO 4:** Panel B voltage (same divider as A).

- **GPIO 5:** Panel B current (same as A).
- **GPIO 6:** Rain sensor analog output.
- **GPIO 7:** Reference mini solar panel across a 10 ohm resistor (reading divided by 10 gives current).
- **GPIO 8 and 9:** I2C SDA and SCL for the optional thermal sensor.
- **GPIO 10:** DS18B20 data (three probes share one wire, 4.7k pull-up to 3V3).
- **GPIO 11:** DHT22 data (10k pull-up to 3V3).
- **GPIO 12:** Tank level sensor.
- **GPIO 13 and 14:** Top and bottom end stops (switches to GND, internal pull-ups).
- **GPIO 15 and 16:** Relay IN1 (pump) and IN2 (spare).
- **GPIO 17, 18, 21:** L298N ENA (PWM), IN1, IN2.
- Avoid GPIO 0, 3, 19, 20, 26-32, 45, 46. Keep all analog readings on GPIO 1-7, because the other analog bank stops working while Wi-Fi is on.

Actuators

- **Pump:** 12 V rail to relay COM, relay NO to pump (+), pump (-) to GND, 1N4007 across the pump (stripe to +). If the relay will not switch off at 3.3 V, drive it open-drain (pin as input means off, output LOW means on).
- **Motor:** L298N OUT1 and OUT2 to the motor, 100 nF across the motor terminals, 220 uF across the L298N 12 V input.

Measurement notes

- ACS712 sits at 2.5 V with no current and the ESP32 analog input is only reliable up to about 2.5 V, so the 4.7k / 10k divider brings its output to about 1.7 V.
- Calibrate against the multimeter. Expect roughly 5-10 % error at 1 A after averaging.
- For fair paper data, run both panels into identical dummy loads during experiments.

9. Experiment plan for the paper

- **Run 1:** both panels clean, no cooling (confirms the panels match).
- **Run 2:** Panel A cooling on, Panel B nothing.
- **Run 3:** Panel A dirty then cleaned, Panel B left dirty.
- **Run 4:** full automatic mode on A, B untouched.

- Record panel temperature, power, soiling index, water used, and the energy the pump and motor consumed.
- The key result is **net gain**: extra energy generated minus energy spent on pumping and moving. Without this, the claim will be questioned.
- A water flow meter (available in the lab) measures water use.
- A FLIR thermal camera (SMARTS lab) can spot-check panel temperatures.

10. Team split and build order

- **EEE:** power path, wiring, fuses, protection, calibration.
- **Mechatronics:** frame, rail and belt, carriage, 3D printed parts.
- **CSE 1:** ESP32 firmware and control logic.
- **CSE 2:** dashboard, database, cloud logging.
- **CSE 3:** data analysis, soiling index, optional camera ML, paper writing.

Build order:

1. Sensors and logging on a breadboard.
2. Calibrate voltage and current.
3. Set the LM2596 to 5.0 V, then connect the ESP32 and relay.
4. Test the pump into a bucket.
5. Print and test one carriage block with a bearing and rod.
6. Assemble rails, belt and motor; test dry with end stops.
7. Add squeegee and spray bar; test cleaning with water.
8. Add battery and charger, move to perfboard and enclosure.
9. Run the experiments.
10. Write the paper; do a patent search before filing.

11. Safety

- Water and electricity share this build: seal connectors, keep the electronics box above the splash zone, use cable glands.
- Always charge the 3S pack through the BMS. Never short or leave the pack unfused.
- Fuse every power branch and keep a master switch within reach.
- Print in PETG or ASA, never PLA, because the panel surface gets hot.

12. College lab availability

All 68 parts were checked against the MDP/IDP lab lists (rooms AB1-213 and AB3-611) and the IoT lab list (AB3-310). Quantities are not in the lists, so confirm counts with lab staff. Some items sit in the IoT lab or SMARTS lab, so check whether you can borrow from there.

Pick up from AB1-213 and AB3-611

- 18650 cells (Dmegs INR18650-26E), 2-cell holders
- LM2596 buck converters x2
- ESP32-S2-DevKitM-1, micro USB data cable
- ESP32-CAM (optional), DHT22
- 25GA-370 gear motor (60 RPM), L298N, 5 V relay module
- MG995 servo (optional), LCD 1602 with I2C (optional)
- Ultrasonic sensor, IR sensor modules x3, push buttons x3, soil moisture sensor
- Multimeter, soldering kit, wire stripper, pliers and tweezers, jumper wires, breadboard

AB1-213 only

- ACS712 30 A x2, water flow meter, container box (tank, if watertight)
- Optical rotary encoder and AMG8833 (optional)

AB3-611 only

- Optical liquid level sensor, mini solar panel (0.45 W), steel box (enclosure)
- Grove MLX90614 thermal camera (optional)

AB3-310 (IoT lab)

- DS18B20 probes x3, LDR module (backup), 100 nF capacitors x10, 220 uF capacitors x2 (at least 16 V)
- Resistors 4.7k, 10k, 100k, 1N4007 diodes x3, MOSFET driver (optional)

SMARTS (AB3-311)

- FLIR E8XT thermal camera, to borrow for testing

Must buy

- 12 V 10-20 W solar panels (the lab′s 0.45 W panel is too small)
- Charge controller, 3S BMS, Schottky diode, fuses, master switch, dummy loads
- Pump, nozzles, spray bar tube, tubing, check valve, filter, squeegee
- Rods, LM8UU bearings, cross bar, GT2 belt and pulleys, aluminium angle, springs
- Wire, terminals, XT60 connectors, perfboard, heat shrink, tilt stand, fasteners
- Screwdriver set, hot glue gun
- The lab′s 3D printing facility is not mentioned in the lists, so check whether one exists

13. Mechanical design of the cleaning carriage

Concept

Two 8 mm smooth rods run along the left and right edges of the panel, about 30 mm above the glass. A printed block with two LM8UU bearings rides on each rod. An 8 mm aluminium cross bar joins the blocks, 13 mm below the rod axis. Two printed wiper arms pivot on the cross bar and hold a silicone squeegee on an aluminium angle against the glass. A GT2 belt on the left side runs from a gear motor at the bottom to an idler at the top, clamped to the drive block. The spray bar is fixed at the top of the panel. Stop sensors at both ends of the left rod set the travel limits.

Key dimensions

- Rod: 8 mm, rod axis 30 mm above the glass, blocks hang about 20 mm outside the panel edge.
- LM8UU bearing: 15 mm outer diameter, 24 mm long.
- Pulley and idler axis: 19 mm above the rod axis, in the same vertical plane as the rod axis.
- Panel size assumed: about 35 x 30 cm (measure yours).

14. 3D printed parts (STL files generated)

Files are in the delivered zip with STL, OpenSCAD source, generator scripts, a README and a preview image. STLs are already rotated for printing.

- **00 fit-test coupon (print first):** bearing pockets 15.0, 15.1, 15.2, 15.3 mm and rod holes 8.0, 8.1, 8.2, 8.3 mm. Pick the snug fit.

- **01 carriage block, drive (x1):** left side, belt groove, prints standing on its end.
- **02 carriage block, idle (x1):** right side.
- **03 belt clamp plate (x1):** four M3 screws.
- **04 wiper arm (x2):** pivots on the cross bar, vertical slots set blade height.
- **05 rod end bracket (x4):** grips the rods, two M4 bolts to the frame.
- **06 motor mount (x1)** and **07 motor clamp strap (x1):** for the 25 mm gear motor.
- **08 idler bracket (x1):** top end, slotted axle for belt tension.
- **09 stop sensor clamp (x2):** clamps on the rod and holds a 32 x 14 mm IR module; mount one at each end, rotated 180 degrees.
- **10 spray bar clip (x5):** snaps onto a 12 mm tube.
- **11 mini panel tray (x1):** holds the 80 x 60 mm reference cell.

Print settings: PETG or ASA, 0.2 mm layers, 4 walls, about 40 % infill.

What was and was not verified

- Verified in software: every mesh is watertight and a single solid; measured hole diameters match the intended sizes to about 0.01 mm.
- Not verified: physical print and fit (hence the coupon), and the OpenSCAD files were not opened in OpenSCAD.
- Assumed and to be confirmed: motor barrel of 25 mm and its shaft size, the IR module size, and equal mounting height for all bases.
- Not yet made: pump and tank mount, ESP32-CAM bracket, DHT22 shield (need the real part dimensions).

15. Files produced in the chat

- **Parts list spreadsheet** (solar_cooling_cleaning_parts_list.xlsx): 68 items with price, supplier and status tracking. It was made before the lab check and the later design changes, so it still shows INA219, V-slot wheels and the older printed-part counts.
- **3D parts package** (solar_cleaner_3d_parts.zip) and its preview image.
- Diagrams shown in the chat: mindmap, physical layout, power circuit, ESP32 wiring, carriage side view and carriage end view.

16. Open items and next steps

1. Print the fit-test coupon and set the bearing and rod hole sizes.
2. Update the spreadsheet with the lab-availability column and the design changes.
3. Measure the motor, IR module, pump and tank, and ask for the remaining printed parts.
4. Confirm quantities with the lab staff and ask about borrowing from the IoT and SMARTS labs.
5. Choose the panel size and confirm the frame dimensions.
6. Buy the must-buy items. Prices and availability were not checked during the chat.
7. Do a patent search (for example on Google Patents) and ask the university IP cell before filing. This is not legal advice.